Mining Rich Data Types

- Mining Text Data
- Mining Spatial-Temporal Data
- Mining Graph and Networks

Mining Spatial-Temporal Data: Properties and Types

- Key Properties in Spatial and Temporal Data
  - Auto-correlation: dependencies and correlations in proximate locations and time, e.g., similar traffic status for adjacent intersections.
  - Heterogeneity: heterogeneous and non-stationary in distribution.
- Types of Spatial and Temporal Data

Data type	Definition	Operation	Examples
Event	at a spatial location and a time point	intersections and similarity/difference	intersection of forest fire and a rainfall returns the overlapping area and time
Trajectory	a path of an object over space and time	finding relationships between two trajectories	transportation (e.g., tracing vehicle), epidemiology, ecology
Point reference data	collected using discrete reference points	reconstructing fields and modeling non-stationary random process	using mobile sensors to collect temperature data in a certain region
Raster data	recording observation data at fixed locations and fixed time	converting a raster to a finer or coarser resolution	estimate traffic volume using data from nearby sensors/cameras

Mining Spatial-Temporal Data: Data Models

- Data Models for Spatial Data
  - Object model: points, lines, polygons (e.g., point ⇒ vehicle, line ⇒ road)
  - Field model: spatial information as a function (e.g., temperature in an area)
  - Spatial network model: graphs for relationship (e.g., road network)
- Data Models for Temporal Data
  - Temporal snapshot model: multiple spatial layers associated with time.
  - Temporal change model: a spatial theme using a start time and the incremental changes (e.g., moving of a vehicle with initial location, speed)
  - Event/process model: multiple events and processes over time